CODE QUEENS · DOMAIN 1 · SDC HACKATHON 2026

NEXTARC

Map your next skill. Your CV becomes a metro map of what to learn next.

CV to roadmap | 2 pilot lines | You stay in control



39%

of workers' existing skill sets are expected to change or become outdated between 2025 and 2030.

A global employer outlook, not a graduate job-loss rate.

THE PROBLEM

"With my current skills, what should I learn next, and why?"

HAS A DIRECTION

Needs to know what to learn next for it.

NEEDS TO EXPLORE

Needs to compare routes before choosing.

Source: World Economic Forum, The Future of Jobs Report 2025, published 7 January 2025. Global employer outlook for 2025–2030.

WHY ONLY TWO ROUTES?

Audience: all fresh graduates.
Pilot: CS fresh graduates, two lines.

Two routes let us check the learning content and the source-to-advice behaviour carefully. Graduates keep the final choice.

B

Backend Development

APIs, data and server-side logic

Q

QA Automation

Designing and automating tests

We will validate the pilot before expanding to more routes.

WHO WE DESIGNED FOR

Two CS fresh graduates, one for each way people arrive: with a route in mind, or still exploring.

J

Jordan

CS fresh graduate

Built a course-booking app with Node.js, Express and PostgreSQL.

Node.js | SQL | REST APIs

Has a route: Backend Development

S

Sam

Software Engineering graduate

Wrote automated UI tests with Selenium and pytest, and logged bugs in Jira.

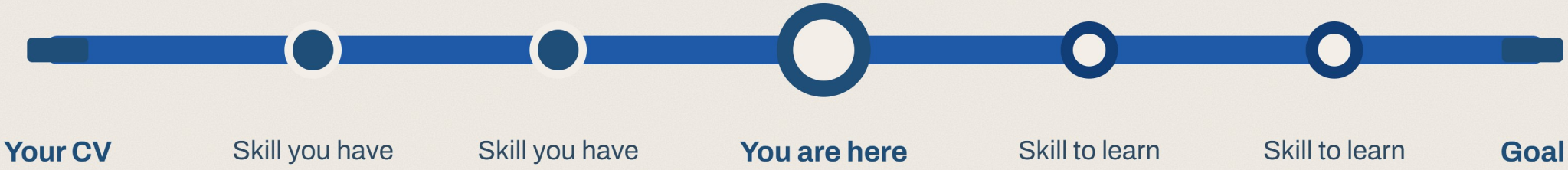
Selenium | Python | Test design

Exploring: unsure which route

Fictional personas and fictional CVs, made for this prototype. Not interview findings.

YOUR CV BECOMES A METRO MAP

Each career route is a line. Each capability is a station. You can see where you are and what comes next.



LINE
a career route

STATION
a capability

FILLED · RING
have it · to learn

INTERCHANGE
a skill on both lines

HOW IT WORKS

Four steps. The AI suggests; you confirm the profile and choose the route.

The ranking never decides for the graduate.

01

Upload your CV

PDF or .txt; the CV file is not saved

02

Check what AI found

Fix, remove or add skills, or mark “not sure”

03

Pick a route, or get suggestions

Explained scores, closest match first

04

Follow your line

Stations to learn next, in order, with reasons

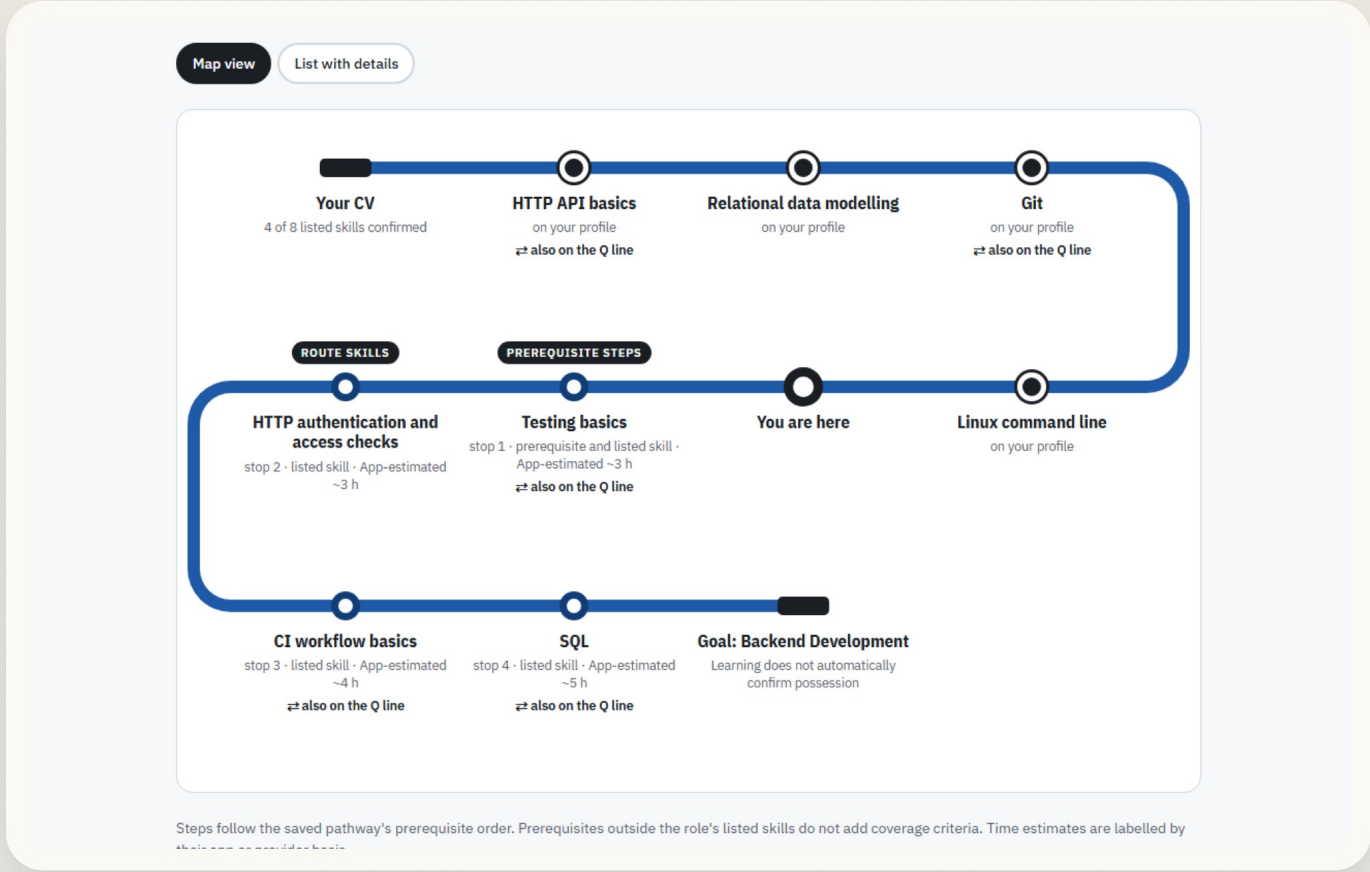
SEE WHERE YOU ARE

01 **Filled stations:** skills on your profile

02 **Learning steps:** prerequisites before dependent steps

03 **Interchanges:** skills on both lines

04 **Open a station:** why it matters, practice time, a starting resource



Screenshot of the live app with a fictional CV. Practice times are app estimates.

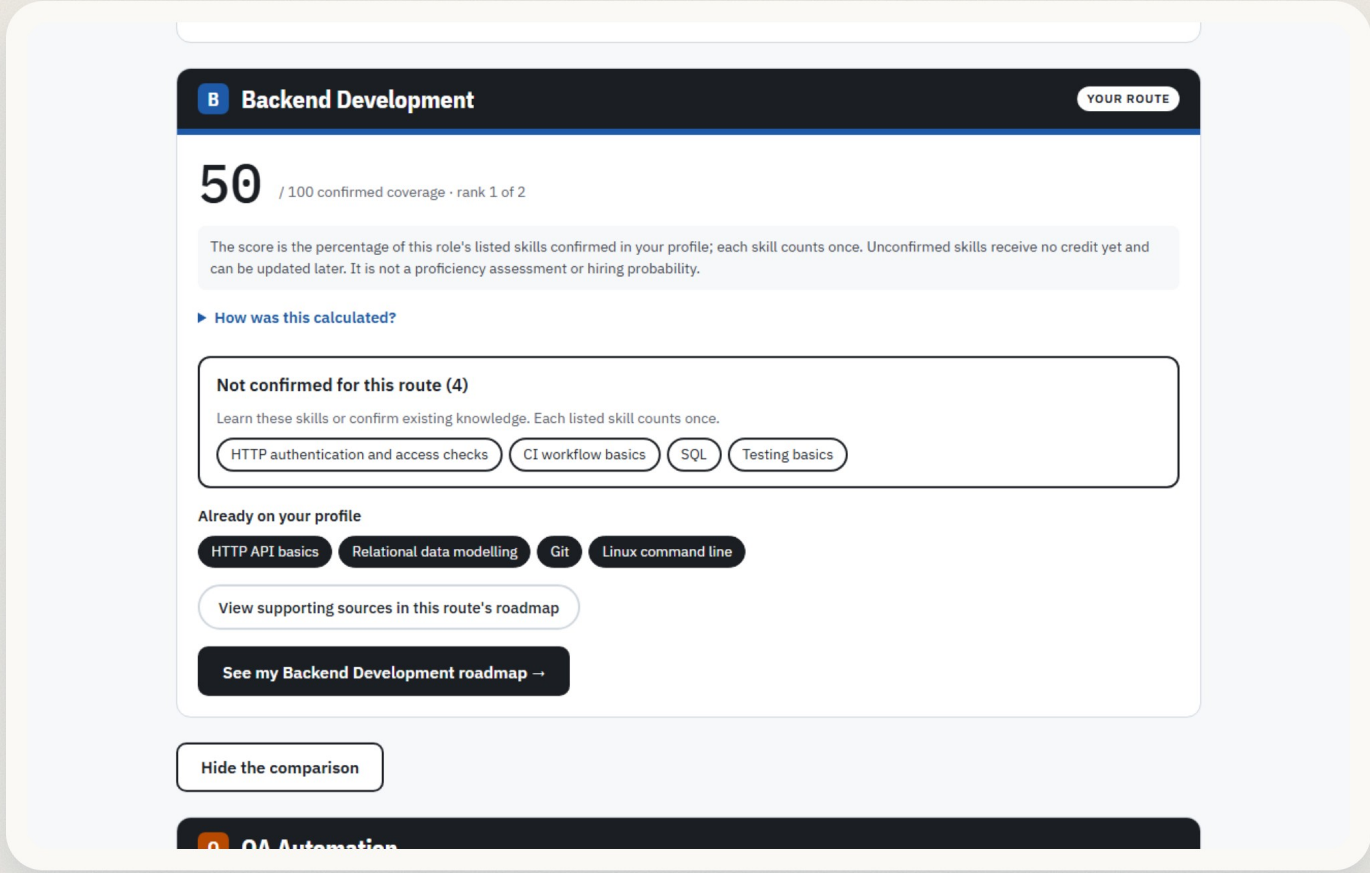
SCORES THAT EXPLAIN

01 **Coverage:** % of the route's listed capabilities you confirmed

02 **Counts once:** each capability, no proficiency bonus

03 **No penalty:** extra unrelated skills never lower it

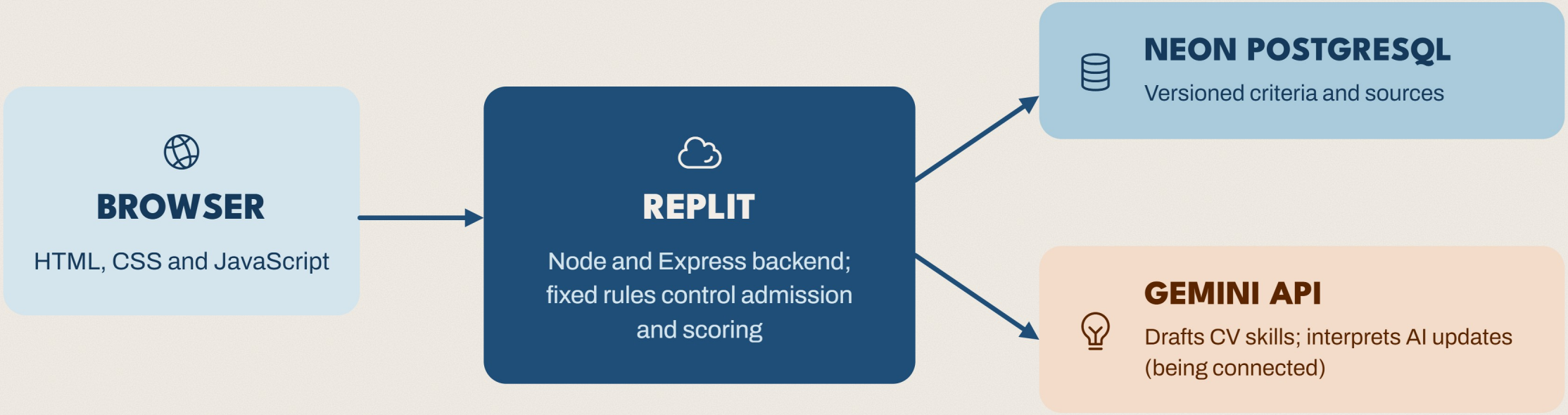
04 **Not a hiring probability;** your chosen route stays yours



Screenshot of the live app with a fictional CV: 4 of 8 listed capabilities confirmed = 50.

ONE APP ON REPLIT

The frontend, the API and the deployment all live in one Replit app.



Matching uses the user-confirmed profile. Our app does not save CV files.

TRUST BY DESIGN

Four rules keep NextArc honest with the graduate.

They are built into the API itself, not just the screens.



AI drafts, you confirm

Only the confirmed profile is used



CV file not saved

Our app processes it in memory



Scores explained

Each capability counts once; not a hiring probability



Fixed rules check AI

AI suggests; fixed rules decide what counts

ADDING A NEW LINE

The engine reuses structured role data. Our tested scope today remains Backend and QA.

01

Validated role scope

Which capabilities the route covers

▶

02

Mappings and content

Task mappings, learning resources, SDG explanations

▶

03

Configuration checks

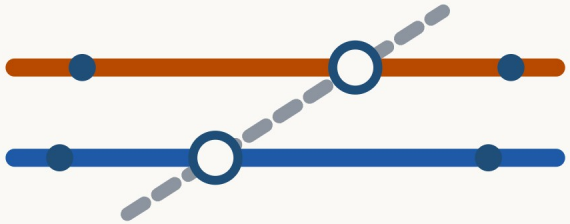
The same fixed rules as today's routes

▶

04

Tested journey

Checked end to end before it goes live



We validate the pilot before expanding. A new line would cross today's lines, showing which skills carry across careers.

TREND EVIDENCE

Automatic source updates, checked by fixed rules before they change any advice.

Status: built and deployed. Switched off for this round to save our free-tier Gemini quota for reading your CV. The app runs on our verified, dated skill lies.

HOW AN UPDATE REACHES ADVICE

1. Feeds: GitHub Changelog and JetBrains AI updates
2. Gemini interprets the retrieved text
3. Fixed rules check support, role relevance and whether it is a distinct capability
4. Versioned criteria and learning guidance

THE RULES

- A new tool name alone adds no scored skill
- Each distinct capability still counts once
- If updates fail, the last valid criteria stay

SDG

4

Target 4.4

OUR PRIMARY SDG

QUALITY EDUCATION

Target 4.4: relevant technical and vocational skills for work. Each role and recommended skill explains how its practice supports technical learning for work.

These are intended contributions, not measured employment results.

Source: United Nations, Goal 4, Target 4.4 (sdgs.un.org/goals/goal4).

ALL SIX FEATURES

Where NextArc stands today, honestly.

Solid tiles work in the live app. Dashed tiles are in integration or planned for the final round.

01

Skills profile intake

Working

02

AI-trend signal layer

Built and deployed; live connection in integration

03

Future-role matching

Working

04

Personalised pathway

Working

05

SDG mapping

Working: Goal 4, Target 4.4

06

Handling uncertainty

Planned: moderate and faster AI-adoption scenarios

WHAT'S NEXT

01

Live trend updates

Finish connecting automatic source updates to the live advice.

02

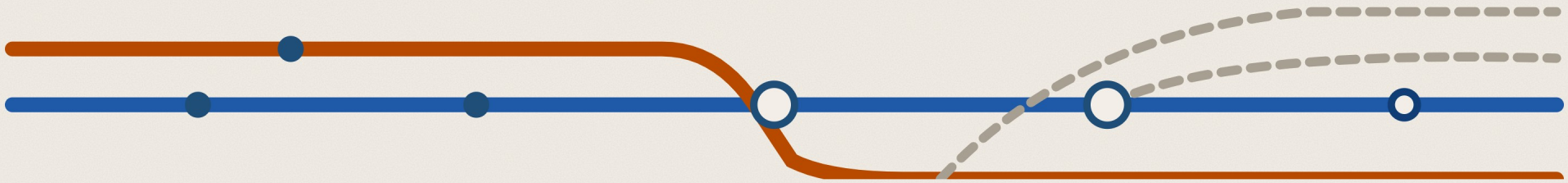
Future scenarios

Moderate and faster AI adoption, with evidence and visible assumptions.

03

Validate, then expand

Validate the pilot before adding more routes.



NEXTARC

YOUR SKILLS. YOUR ROUTE. YOUR NEXT STEP.

Chew Khai Shuen
Team leader, trend evidence, docs

Teoh Chia Sin
Frontend, design, pitch, video

Jeremy Tey Jie Ming
Backend and deployment

Gan Yi Jong
Trend evidence and docs

sdc-hackathon-code-queens-1.replit.app · github.com/khaishuen819/SDC_Hackathon_CodeQueens